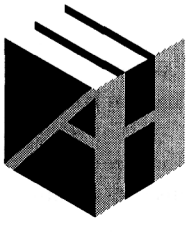
Digital Processing of Synthetic Aperture Radar Data

Algorithms and Implementation

Ian G. Cumming
Frank H. Wong



ARTECH
HOUSE
BOSTON | LONDON
artechhouse.com

Contents

Foreword xix

Preface xxiii

Acknowledgments xxv

I Fundamentals of Synthetic Aperture Radar 1

1 Introduction 3

1.1 Brief Background of SAR 3

1.2 Radar in Remote Sensing 5

1.3 SAR Fundamentals 5

1.4 Spaceborne SAR Sensors 10

1.5 Outline of the Book 14

1.5.1 Example of a Spaceborne SAR Image 16

References 19

2 Signal Processing Fundamentals 21

2.1 Introduction 21

2.2 Linear Convolution 22

2.2.1 Continuous-Time Convolution 22

2.2.2 Discrete-Time Convolution 24

2.3 Fourier Transforms 27

2.3.1 Continuous-Time Fourier Transform 27

2.3.2 Discrete Fourier Transform 28

2.3.3 Fourier Transform Properties 30
2.3.4 Examples of Fourier Transforms 34
2.4 Convolution Using DFTs 36
2.5 Sampling of Signals 38
2.5.1 Spectrum of Sampled Signals 38
2.5.2 Signal Types 40
2.5.3 Nyquist Sampling Rate and Aliasing 42
2.6 Smoothing Windows 49
2.7 Interpolation 51
2.7.1 Sinc Interpolation 52
2.7.2 Spectra of Interpolation Kernels 56
2.7.3 Nonbaseband and Complex Interpolation 58
2.8 Point Target Analysis 59
2.9 Summary 64
2.9.1 Magellan Image of Venus Crater 65
References 66

3 Pulse Compression of Linear FM Signals 69
3.1 Introduction 69
3.2 Linear FM Signals 70
3.2.1 Time Domain Representation 70
3.2.2 Spectrum of the Linear FM Pulse 72
3.2.3 Sampling of FM Signals 76
3.2.4 Frequency and Time Discontinuities 78
3.3 Pulse Compression 80
3.3.1 Principles of Pulse Compression 80
3.3.2 Time Domain Compression of Linear FM Signals 82
3.3.3 Frequency Domain Matched Filter 87
3.3.4 The Effect of Windows 90
3.3.5 Oversampling Ratio Revisited 93
3.4 Matched Filter Implementation 93
3.4.1 Target Registration and Matched Filter Throwaway . . 96
3.5 FM Rate Mismatch 98

3.5.1 Effect of Mismatch in a Baseband Signal 98
3.5.2 Effect of Mismatch in a Nonbaseband Signal 102
3.5.3 Filter Mismatch and Time Bandwidth Product 102
3.6 Summary 103
3.6.1 A Wide Swath ENVISAT/ASAR Image 104
References 107
Appendix 3A: Derivation of the Matched Filter Output 108
3A.1 Signal and Matched Filter of Equal Length 108
3A.2 Signal and Matched Filter of Unequal Length 110
Appendix 3B: Derivation of the Phase Error Due to Mismatch . . . 111

4 Synthetic Aperture Concepts 113
4.1 Introduction 113
4.2 SAR Geometry 114
4.2.1 Definition of Terms 114
4.2.2 Satellite Ground Range Geometry 120
4.2.3 Satellite Orbit Geometry 122
4.3 The Range Equation 125
4.3.1 Hyperbolic Form of the Range Equation 125
4.3.2 Relationships Between Velocities and Angles 128
4.4 SAR Signal in the Range Direction 130
4.4.1 Transmitted Pulse 130
4.4.2 Data Acquisition 131
4.5 SAR Signal in the Azimuth Direction 133
4.5.1 What Is Doppler Frequency in the SAR Context? . . . 133
4.5.2 Coherent Pulses 134
4.5.3 Choice of PRF 136
4.5.4 Azimuth Signal Strength and Doppler History 137
4.5.5 Azimuth Parameters 140
4.6 The Two-Dimensional Signal 141
4.6.1 Data Arrangement in Signal Memory 142
4.6.2 Demodulated Baseband Signal 144
4.6.3 The SAR Impulse Response 146

4.6.4 Typical Radar Parameter Values 147
4.7 SAR Resolution and Synthetic Aperture 147
4.7.1 Resolution Derived from Bandwidth 147
4.7.2 Synthetic Aperture 151
4.8 Summary 153
4.8.1 ScanSAR Narrow Image of Vancouver Island 155
References 156
Appendix 4A: Derivation of the Approximate Radar Velocity 157
Appendix 4B: Quadrature Demodulation 159
4B.1 Theory of Quadrature Demodulation 160
4B.2 Errors and Corrections 161
Appendix 4C: Concept of Synthetic Aperture 164

5 SAR Signal Properties 169
5.1 Introduction 169
5.2 Signal Spectrum in the Low-Squint Case 170
5.2.1 Spectrum in the Range Doppler Domain 171
5.2.2 Two-Dimensional Spectrum 172
5.3 Signal Spectrum in the General Case 172
5.3.1 Range Fourier Transform 174
5.3.2 Azimuth Fourier Transform 175
5.3.3 Range Inverse Fourier Transform 178
5.4 Azimuth Aliasing and the Doppler Centroid 182
5.4.1 Origin of Azimuth Aliasing and Ambiguities 182
5.4.2 The Doppler Centroid 185
5.4.3 The Doppler Ambiguity 188
5.4.4 Doppler Centroid Variation with Range 189
5.5 Range Cell Migration 194
5.5.1 RCM Components 195
5.5.2 Multiple Targets at the Same Range 198
5.5.3 Target Trajectory Wraparound 199
5.6 Point Target Examples 200
5.6.1 Simulation Parameters 200

5.7 Prelude to SAR Processing Algorithms 204
5.7.1 Time Domain Matched Filtering 205
5.7.2 Real-Time Processed Airborne Radar Image 207
5.7.3 Unfocused SAR 209
5.7.4 Moving on to Better Processing Algorithms 210
5.8 Summary 211
References 214
Appendix 5A: Range/Azimuth Coupling 215
Appendix 5B: A Note on the Azimuth FM Rate 219
5B.1 Geometric Interpretation from Slopes 220
5B.2 Approximate Signal Representation 221

II SAR Processing Algorithms 223

6 The Range Doppler Algorithm 225
6.1 Introduction 225
6.2 Algorithm Overview 226
6.3 RDA in the Low Squint Case 229
6.3.1 Raw Radar Signal Data (Raw Data) 229
6.3.2 Range Compression 231
6.3.3 Azimuth Fourier Transform 232
6.3.4 Range Cell Migration Correction 235
6.3.5 Broadening Due to Residual RCM 241
6.3.6 Azimuth Compression 243
6.3.7 Example of a Low Squint RADARSAT-1 Image 248
6.4 The High Squint Case 250
6.4.1 Modifications to Handle Squint 250
6.4.2 Implementation of SRC 253
6.4.3 SRC Options Needed for Satellite and Airborne Cases . 256
6.4.4 SRC Simulation Experiments 258
6.4.5 An L-Band Airborne Radar Image Example 264
6.5 Multilook Processing 265
6.5.1 Frequency and Time Relationship of Looks 265

6.5.2 Look Extraction, Detection, and Summation 267
6.5.3 Equivalent Number of Looks 270
6.5.4 An Example of Multilook Processing 271
6.5.5 FM Rate Error 276
6.5.6 Multilook Processed Image 278
6.6 Summary 280
References 281

7 The Chirp Scaling Algorithm 283
7.1 Introduction 283
7.1.1 Overview of the Chirp Scaling Algorithm 284
7.2 The Chirp Scaling Concept 287
7.3 Applying Chirp Scaling to RCMC 294
7.3.1 Bulk and Differential RCMC 295
7.3.2 A More Accurate Expression for RCM 298
7.4 Derivation of the Scaling Function 300
7.4.1 Examples of the Size of Differential RCM 302
7.5 CSA Processing Details 304
7.5.1 Range Processing 304
7.5.2 Azimuth Processing 309
7.6 Processing Examples 310
7.6.1 Simulations with Point Targets 310
7.6.2 Processing of SRTM/X-SAR Data 315
7.7 Summary 316
References 319

8 The Omega-K Algorithm 323
8.1 Introduction 323
8.1.1 Overview of the Omega-K Algorithm 324
8.2 Reference Function Multiply 328
8.3 Stolt Interpolation 330
8.3.1 The Change of Variables 330
8.4 Interpretations of the Stolt Mapping 335
8.4.1 Components of the Stolt Mapping 335

8.4.2 Interpretation Via Fourier Transform Properties 336
8.4.3 Interpretation Via the Region of Support 340
8.4.4 Interpretation Via Imaging Geometry 341
8.5 Error Analysis 343
8.6 Approximate Version of the Omega-K Algorithm 345
8.6.1 Approximation Components 346
8.6.2 Relation to the RDA and the CSA 348
8.6.3 Discussion of Errors in the Approximate ωKA 348
8.7 Processing Examples 349
8.7.1 Simulation of the Full ωKA Algorithm 349
8.7.2 Approximate Version of the ωKA 356
8.7.3 An X-Band Airborne Radar Spotlight Image Example . 357
8.8 Summary 358
References 359
Appendix 8A: Stolt Mapping in the Wavenumber Domain 362

9 The SPECAN Algorithm **369**
9.1 Introduction 369
9.1.1 Overview of the SPECAN Algorithm 370
9.2 Derivation of the SPECAN Algorithm 372
9.2.1 From Convolution to SPECAN 372
9.2.2 Geometric Interpretation 373
9.2.3 Aliasing and FFT Length 377
9.2.4 Output Sample Spacing 379
9.2.5 Number of Good FFT Output Points 380
9.2.6 Placement of Subsequent FFTs 382
9.2.7 Stitching the FFT Outputs Together 385
9.3 Multilook Processing 385
9.4 Processing Efficiency 389
9.5 Range Cell Migration Correction 393
9.5.1 Time Domain Linear RCMC 394
9.5.2 Skewing and Deskewing 394
9.6 Phase Compensation 396

9.7 Image Quality Issues 400
9.7.1 Frequency Discontinuities at the Stitching Points 400
9.7.2 Azimuth FM Rate Error 403
9.7.3 Radiometric Scalloping 405
9.8 Processing Examples 416
9.8.1 Simulated Point Targets 416
9.8.2 ERS Image Processed by SPECAN 418
9.9 Summary 420
References 421

10 Processing ScanSAR Data **425**
10.1 Introduction 425
10.2 ScanSAR Data Acquisition 427
10.3 Compression of a Single-Burst Target 431
10.4 Full-Aperture Processing 434
10.5 The SPECAN Algorithm 437
10.6 The Modified SPECAN Algorithm 438
10.6.1 Outline of the Algorithm 438
10.6.2 SRTM Processing Example 442
10.7 The Short IFFT Algorithm 442
10.8 The Extended Chirp Scaling Algorithm 447
10.9 Stitching Processed Bursts Together 450
10.10 Summary 453
10.10.1 RADARSAT-1 ScanSAR Image 454
References 455

11 Comparison of Algorithms **461**
11.1 Introduction 461
11.2 Recap of the Precision Processing Algorithms 461
11.2.1 Range Doppler Algorithm 461
11.2.2 Chirp Scaling Algorithm 462
11.2.3 Omega-K Algorithm 462
11.3 Comparison of Processing Functions 463
11.3.1 Form of Range Equation 465

11.3.2 Implementation of the Azimuth Matched Filter 465
11.3.3 Implementation of Range Cell Migration Correction . . 465
11.3.4 Implementation of Secondary Range Compression . . . 466
11.4 Summary of Processing Errors 466
11.4.1 QPE in the Azimuth Matched Filter 467
11.4.2 QPE in Secondary Range Compression 468
11.4.3 Residual RCM 469
11.4.4 Examples of the Size of the Processing Errors 470
11.5 Computation Load 473
11.5.1 Basic Algorithm Operations 473
11.5.2 Range Doppler Algorithm 473
11.5.3 Chirp Scaling Algorithm 474
11.5.4 Omega-K Algorithm 475
11.6 Pros and Cons of Each Algorithm 476
11.6.1 Pros and Cons of the RDA 476
11.6.2 Pros and Cons of the CSA 477
11.6.3 Pros and Cons of the ωKA 478
11.7 Summary 479
11.7.1 ASAR Image of the Strait of Messina 480

III Doppler Parameter Estimation **481**

12 Doppler Centroid Estimation **483**
12.1 Introduction 483
12.1.1 Doppler Centroid Frequency 483
12.1.2 Satellite SAR System Geometry 484
12.1.3 Outline of This Chapter 487
12.2 Doppler Centroid Accuracy Requirements 489
12.2.1 Baseband Centroid Accuracy Requirements 489
12.2.2 Doppler Ambiguity Accuracy Requirements 497
12.3 Calculating Doppler Centroid from Geometry 497
12.3.1 Examples of Doppler Centroid Calculations 501
12.3.2 Yaw and Pitch Steering 503

12.4 Estimation of the Baseband Centroid from Received Data 504
12.4.1 Magnitude-Based Estimation Approach 504
12.4.2 Phase-Based Estimation Method 509
12.5 Estimation of the Doppler Ambiguity from Received Data 514
12.5.1 A Magnitude-Based DAR Estimation Method 515
12.5.2 Phase-Based DAR Estimation Methods 518
12.5.3 The Wavelength Diversity Algorithm 521
12.5.4 Multilook Cross Correlation Algorithm 526
12.5.5 Multilook Beat Frequency Algorithm 530
12.5.6 PRF Diversity Method 535
12.5.7 Comparison of DAR Algorithms 536
12.6 Principles of the Global Estimation Procedure 539
12.6.1 Spatial Diversity 539
12.6.2 Estimator Quality Measures 540
12.7 Surface Fitting Approaches 543
12.7.1 Global Fit Using a Polynomial Surface 543
12.7.2 Global Fit Using a Geometry Model 545
12.7.3 Automatic Fitting Procedure 546
12.8 Summary 549
References 550
Appendix 12A: Details of the Doppler Calculation 553
Appendix 12B: The Offset Frequency in DAR Algorithms 561
12B.1 Model of Signal from Each Look 561
12B.2 Multilook Cross Correlation Algorithm 562
12B.3 Multilook Beat Frequency Algorithm 565

13 Azimuth FM Rate Estimation **567**
13.1 Introduction 567
13.2 Azimuth FM Rate Accuracy Requirements 567
13.3 Geometry Models to Compute the Azimuth FM Rate 570
13.4 Estimating the Azimuth FM Rate from Data 574

13.4.1 Autofocus Via Contrast Maximization 575
13.4.2 Autofocus Via Look Misregistration 576
13.4.3 Phase-Based Autofocus Methods 582
13.5 Summary 584
13.5.1 A Miniature SAR System – MiSAR 584
References 586

Appendix A: RADARSAT Data CD **587**

List of Acronyms **589**

List of Symbols **593**

Selected Bibliography **597**

About the Authors **617**

Index **619**